[00:00] 发言人 1: After learning these two concepts, let’s put them side by side so you have a better idea. Okay, what it does. So if you.

[00:10] 发言人 1: Take a look at all the left something you immediately left versus right you need anything you notice that all of these possible states only occur once at the left in the page. But these notes can happen many times on the surgery depending on the particular route you take. Later on you will see why.

[00:33] 发言人 1: So another thing you notice is that every note on the right of your surgery actually corresponding to a particular path in the space space. Okay， so let’s take this G for example。Let’s say now you are looking at this particular node on your search space，on your search tree，and what it does is that if you actually trace it back。你 actually. But they actually cause loss to this.

[01:13] 发言人 1: red curve on the left 然后啊 red ok so let’s say now you can take another number for example let’s say i take this q. 我们。Unique right, you only once you want to put the direction back to its group.

[01:32] 发言人 1: So you do a face side to this space space craft. What it does is you find your field right here. 你没听。Pointing to this bus and.

[01:46] 发言人 1: Right here and the E, right here and the C, right here and the B. So this known Q. Basically corresponding to this typically yellow path.

[02:02] 发言人 1: Okay，and even if you have the same no。For example，this。了解剩余的功能。But they occur in different locations of your search because they actually take a different path to reach this time.

[02:23] 发言人 1: So if you trace that. It goes to F to R to E and back to S. Okay, so again, in this state space graph, this is unique.

[02:35] 发言人 1: So you from here you go there, this is this part right here, right? Then F to R. THIS PART RIGHT HERE. Then R to F, R to F right here.

[02:50] 发言人 1: And RPD，so RPD right here，and it is right。THIS IS THE COST WE ARE LOOKING AT TO REACH THIS BOOK. Okay，so I hope this gave you a clear idea about the correspondence between state space graph versus surgery。

[03:09] 发言人 1: Okay，so state space graph basically captures the purposes that it captures the whole dynamics of the problem by showing you what are all the unique states that possible and how you can go from one to the other。So here he tells you, ask cannot direct people to be. 或者。

[03:29] 发言人 1: Okay，and the surgery is basically a tentative plan。Most likely maintained by a search algorithm about what are you supposed route that you want to take. ok. Any question on this one?

[03:48] 发言人 1: Okay，another important thing is like I repeat already，both of these search rates，the search tree and space space graph，we only build。As little as possible. We never build a whole thing into that.

[04:03] 发言人 1: Okay, we just still as much as see. Okay, so now if we learn these two concepts in mind, we can actually try this out on any search problem. And later on,

you will see that everything goes search out with only differentiate in the main how they come out the surgery.

[04:22] 发言人 1: Okay，so this example we have talked about is actually borrowed from the AI Facebook。So this problem is called traveling in Romania，you have this road，and what we want to do is we want to go from Iraq，so this is our starting node S，and we want to arrive at this Bucharest，this is our goal node D。Okay，and we want to find around that can take you from message。

[04:49] 发言人 1: okay，so. If you try thinking about this problem and put this problem into the concept of surgery as we are talking about，this particular graph you see over here，takes this things from the graph。Okay，then what is the sort of tree head for this product?

[05:09] 发言人 1: Okay，so basically if you want to solve this problem，it will require basic three steps. Okay，the first step is you want to expand this particular class. For example，if you are starting to say starting over on.

[05:25] 发言人 1: 你立刻知道什么是 possible designation，你可以去。And all these potential plans we call the pieces in a place we call the frame. Okay, this frame comprises basically keep track of the potential plans that’s currently under your consideration.

[05:44] 发言人 1: Okay, in a few slides later we will tell you exactly what this frame means. And for this week’s tutorial we actually provide you with a high level of dedication of this frame. So you know exactly what this print actually does.

[05:58] 发言人 1: Okay. So now, if you take a look at this, then you can basically get, we can basically put every three search algorithm into the following canonical schematics. Okay, any algorithm, any search algorithm can essentially be factored.

[06:17] 发言人 1: If any three-based search algorithm can be factored using the following scheme. Okay, so this commerce demo basically requires you to solve a problem and a strategy that you want to expand and explore. Okay, so you will see that this strategy basically defines different kind of search algorithm.

[06:40] 发言人 1: For example， the death first， breath first， uniformity， a star algorithm， basically only different in the strength， nothing else。Okay，and you are always starting the stream with the initial state of the problem. Okay，then you loop in the end.

[07:00] 发言人 1: You either find a solution, the solution remember any solution to a certain problem is basically a sequence of actions that take you from the start to the goal. Okay, so either you find this solution, the sequence of actions, or you do not find so you declare it. Okay, then as you are working the problem, you need to keep track of dependent plan okay and you need to decide a way how you expand all these trees.

[07:35] 发言人 1: For example if you take a look at here, if you start from the rod, then you can go to thisville, you can go to this p city and the z cities, then you need to decide which one to go at and how do you calculate all. In what secret right now? Okay, and this is what we call the exploration strategy basically which one visit?

[07:56] 发言人 1: Okay, so all these different elements of this. Only going to be different in this particular strategy, how you populate, how you expect of this. Okay, so let’s actually take the following example.

[08:11] 发言人 1: Okay, to understand exactly what the previous this suitable actually means. Okay. Again, back to this very tiny search example.

[08:22] 发言人 1: So you’re remembering a search tree now. This is the space space graph right now. We want to search, we want to go from S to G.

[08:31] 发言人 1: So by doing so, we may take our search tree. The first thing we remember from the search tree is we always start with starting your S. Right and to the right.

[08:43] 发言人 1: is the actual surgery that we implement。Okay，and you can consider this as. The bridge.

[08:57] 发言人 1: Okay, so from this starting state, what you can do is you can either go to D. Or you can go to B, or you can go to P. So these three are currently. Plans under your consideration.

[09:21] 发言人 1: Right, then if you go back to the search algorithm, once you populate the current plans on the consideration, the next thing you need to choose is you need to have a strategy, apply the strategy to choose one of them. Okay, so let’s say in here we just... So the next we will show you three different elements on each one have a different strategy. But for now, let’s say we have some strategy and we are going to pick one of them.

[09:50] 发言人 1: Let’s say we pick B. Okay, so if you go back to the last. 这个目标，他 says we need to now。Insert whatever node we have expanded to achieve.

[10:04] 发言人 1: So what it means is this. So let’s say now you going from the S node, you look at you can go to D, E, or P. Right? Now you want to go to P, or you choose P for some strategy.

[10:17] 发言人 1: Okay, this is why you. The thing you do is you maintain a queue. So if you have learned data structure, basically the queue is something you first put in the last come out.

[10:31] 发言人 1: Okay, so now you put this s. 嗯，你 put the note into it after you have move on to his challenge。So we put s in after we select b.

[10:48] 发言人 1: As this child knows the truth. Okay, then after you looking at the speed, you can see for B, you can go to B, C and E. So from the.

[11:00] 发言人 1: These B C and E are for the plans that under your considerations. Okay，so all of these B C and E。You apply some strategy and you choose a node to visit.

[11:13] 发言人 1: Let’s say for some strategy, you choose help you to choose the E you want to visit. And as soon as you visit E, you put its parent node into this queue. So originally you only have S, now you move on to E, so its parent node of E is in queue.

[11:32] 发言人 1: Okay，now let’s say you move on to E from E，you can either go to H or R，and H and R is the two that’s currently under consideration，along with

the previous one that you have visited, that you have not completely filled。Okay, now from E, you can either go to Asia or go to R, then there's some strategy tells you, let's say now the strategy tells you to move on to R。So if you look at the queue, you already have asked.

[12:00] 发言人 1: then after you go to you go to b you put this parent hands in here you go to e you put this parent d over here you visit r you put the art parent e into the queue so now this is your queue. Okay, now from R you can only go to F, so there's no question you can only move to R, so after you moving to R。You put this parent E into the queue, so in your queue you have S D E.

[12:33] 发言人 1: AND R. Okay, now from there you can either go to C and G, right? You can either go to C or G。

[12:44] 发言人 1: Let's say out of some strategy he asks you to go to G, then as soon as you go to G, you put the G's character in here, so you have S。You have D, you have E, you have R, you have F. That's what's currently in your queue, right?

[13:06] 发言人 1: Then now you visit no G and you do a test if you see over here. You always need to test if this is a goal state. Okay, if this is goal state, then you return this goal state.

[13:22] 发言人 1: Okay, so after you move on to the node G, you do a test and you realize this is your goal. That's G, then you... Then you put this G in here as well. Okay, so first of all, something you notice is that this is your space space graph, and this is the surgery that we have been calculating as we try to go from SG。

[13:50] 发言人 1: The first thing you notice is that. We only explore part of the tree, all these part we have not visited. Even though they exist, and many of these we actually did not visit as well.

[14:07] 发言人 1: Okay, thank all. Okay, we have never visited. Sorry, another thing is remember we have a TV queue。

[14:20] 发言人 1: The algorithm asks us to keep a queue, right? It just be be. 嗯。R and G, so why we want to keep such a queue?

[14:34] 发言人 1: Is when the whole algorithm stops, we can easily pop the entries from the queue, and whatever we pop out on the queue basically is the solution we found. Let's say now we have this queue, the link that we have, the queue, the element that we pop out from this is safety. G R F E D.

[15:00] 发言人 1: Yes, this is basically the solution we found, the sequence of actions that take you from a start to goal. Okay, so maintaining this Q data structure helps us to keep track of all these. Okay, and this is basically the fringe we are talking about.

[15:19] 发言人 1: All the plans that's currently under consideration. Okay, any questions on this part? Okay, hope that's clear enough。Now with this in mind, we can talk about specific search algorithms。Okay, like I said, they only going to differ in the way how they choose。

[15:40] 发言人 1: For example, if you have DDP, which one shall I choose? If I have CDE, which one shall I choose? That's the only difference about all the search

algorithms that based on this。Okay, so now before we look at the specific details of each one of these algorithms, let’s look at.

[16:00] 发言人 1: What are the properties? What are the interested properties of a tree search of any search object? Okay， the first property that we want to investigate is, let’s say the search problem exists，a solution，so there’s a solution exists，you know this problem is solvable in some sense。Can your algorithm guarantee the fight? And this.

[16:28] 发言人 1: Generically, you know, has completeness of the search algorithm. Is your algorithm complete? Means that if the problem you’re trying to solve has a solution, a solution is your algorithm going to guarantee to find it.

[16:44] 发言人 1: It’s called the completeness of the search algorithm. The other concept sometimes you hear more often is called optimal melody or optimal is your search algorithm up finding the optimal solution so what it means is that let’s say. This problem are probably want to solve has a solution, and maybe multiple solution.

[17:06] 发言人 1: Like in traveling in Romania, just like this small search problem we showed you, there are multiple solution, there are multiple sequence of action that can take you from the start to S to G. What is your algorithm guarantee to provide you with the shortest path? that can take you from the start to the goal.

[17:31] 发言人 1: okay, so let’s say what this optimality concern is. the completedness tells you if the solution exist, you are going to find it. become optimality tells you.

[17:43] 发言人 1: The solution you return is going to be off. Okay, so basically these are concerning the quality. observation. 他说。Okay，you have an algorithm。

[17:59] 发言人 1: The first thing you care about is the quality of the outcome，the solution found by your algorithm。So in addition to the quality of the solution that you care, in reality we equally care about the complexity of the algorithm itself, meaning that how complicated or how long it took for your algorithm to actually solve. so there’s basically two complexities that we may be interested one is called time complexity.

[18:31] 发言人 1: Okay，you can basically imagine this to be the time is not necessary the physical time like one hour thirty minutes because that depends on the hardware capability the time complexity here more than these the ultimate complexity like how many operations how many time how many calculations your algorithm need to perform. to actually solve it okay another complexity. It’s called space complexity.

[19:02] 发言人 1: So when you’re solving a problem, how big is the French? So this... How big is the tennis plan you need to keep? In order to solve your problem, okay?

[19:18] 发言人 1: So some algorithm only requires small piece of the memories, you know? Some algorithm require a lot, okay? So these two are more concerned about how efficient your algorithm itself to find a solution.

[19:32] 发言人 1: The first two more concerned about the solution itself. Okay，so now if we try to understand。These are time complexity and space complexity.

[19:43] 发言人 1: What we do in this class is we are using this small cartoon. Ok, so imagine this cartoon is basically the search tree. You will always start with one node, the starting node, and you are starting to popular.

[19:59] 发言人 1: Right, so. At every part of this node, let’s say we decide a hypothetical number called the branching factor. Okay, so what it means is that at any given state.

[20:13] 发言人 1: What are the maximum number of children that you can visit the space be the breath? Okay, let’s say in the previous case, if we take this as a company example. But any note in this space spacecraft what is the maximum number of challenge of any official.

[20:35] 发言人 1: I think if my eye bowling is right, the number is three. The maximum number of child know that you can visit by any given states in this problem is three. So for example, what else you can go to three? What he can go to three different places. Okay, I think imagine this number to be the right。Okay, so for any node in your space space graph at maximum you can business.

[21:04] 发言人 1: Okay, you can imagine this to be maximum all the expectation they have, both are fine。Maybe imagine this to be the maximum would be better, okay? And the M is the maximum depth of the circuit, okay? So you can go to many different routing of space, base throughout the maximum depth, the maximum number of iterations you can take。okay that means.

[21:28] 发言人 1: The search problem has an end. You do not have a loop in a search tree where you can just looping and those not weapons. It happens for no problem, but let’s assume this problem find at the end of that.

[21:41] 发言人 1: Okay, so now the maximum number of steps that you can take。But any of these passes he was a tree, okay? Then.

[21:54] 发言人 1: Another thing I want you to be aware is that the actual solution can happen in any arbitrary. Yes, right. It can actually happen to the here and it happens to here.

[22:05] 发言人 1: So if you see this search tree before, I think it’s more clear to you. The goal states G over here can happen at depth one, two, three, four, five. If it depth equal to five, it can also happen at depth equal to four.

[22:27] 发言人 1: 在最后。Okay, so your goal can actually happen at different depths depending on the positive。Okay, so this basically corresponding to this cartoon where the depth can happens at different, the optimal solution happens at different depths。

[22:42] 发言人 1: Okay, so if you’re now trying to look at the complexity of this problem, you can basically try to sum up with all the notes in the search tree。Okay, this is way to give you a worst case scenario。好，OK，let’s say if your girlfriend happen to be this。

[23:02] 发言人 1: Okay, then in the end you might explore the whole third tree to find。Okay, so if you sum at the first level you have one node, second level you have B notes because that node can maximum branch to be, you only have one branch,

then in the third level you have B notes, every node at most can branch to be。So you have B into B squared in the last layer.

[23:29] 发言人 1: Let’s say you have maximum N in the end, you have B raised to that. So it’s in the order of B over B raised to power. okay, so if you talk some numbers。If your branch factor is three, like in the previous case, and now if you go to level ten right now, the maximum you go to five or even six, this number is already pretty big.

[23:57] 发言人 1: Okay, so this are the properties that we can use to describe different elements. So we saw this in minds, we can actually looking at three。What we call only for the algorithm again, the only difference in the way how they choose a algorithm。

[24:16] 发言人 1: Okay, so let’s first look at the search called that’s first。Okay, as the name suggested, you will always expand。The deepest node first.

[24:30] 发言人 1: Okay, so this is basically the input strategy to the previous schema, the pseudocode of our research algorithm. Okay, so let’s now let me walk you through this algorithm to actually help you understand. I think it’s straightforward enough, but let’s just go through this one so it becomes easy.

[24:50] 发言人 1: So remember the goal is to start with. Starting over s and we want to go to this technology okay now for this. So as you can either go to D, go to E or go to P right? You can go to D, go to E, go to P okay? And according to this strategy, I’m going to always explore the node with the deepest steps first.

[25:19] 发言人 1: Okay, so if you look at the D E and the P, they all have that equal to one, right? They are similar that。And what happens? What happens is you need to now fix the convention.

[25:33] 发言人 1: So if you have a tie in the depths, which one you want to visit? Let’s say for our case, if we have a tie among solutions about the child, we will always explore the leftmost or the one with a smaller alphabetical number. So let’s say D, E, and P, they all have depths equal to one.

[25:53] 发言人 1: I’m going to pick the one on the left. That’s just some convention we have to fix. Great, OK。

[26:01] 发言人 1: So now we going to D after going to D from D we can go to D C and E and previously from S we can go to E and T so currently we have five note of the consideration. Right. So out of these five notes, the strategy tells me to find the one with the deepest.

[26:26] 发言人 1: So B, C, and D have depth equal to two. And E and P has done even the one. So out of these I would definitely choose one from B, C and E. So now since they are tied, so according to my convention I will choose the leftmost first.

[26:46] 发言人 1: So now I move B, as soon as I move to B, A becomes available, right? From B I can go to A and C and E is still available from B, I have already visited. And from S I visited, if I can go to E and P as well.

[27:05] 发言人 1: So these two have depth equal to one, these two have depth equal to two, and a have depth equal to three. Now according to my strategy, things become easy. I pick the one with the it’s the depth, so I go to the.

[27:20] 发言人 1: Okay，now once I go to A，I have nowhere to go，and I take a check if A is my goal，it is that’s the goal I’m looking for，the answer is no，so I need to look at again，now if I look at my fridge，the current notes that under consideration，I only have C E E P。Right, so out of these four again, these two are that’s one, these two are that’s two. So we have a tie, so I always go my convention, always take the one on the left.

[27:54] 发言人 1: So I’m going to page C once I go to C. The current node that’s under consideration at these four, this part I’ve already visited. So I have A, I can come from C and E, I can come from D already visited by S, I can go to E and P.

[28:12] 发言人 1: So I have to know that that’s one one know that two one know that three according to this strategy, I will always take the one with biggest, so I go to A. So I move on to A, I do a test because no A I cannot go anywhere else. This is not my goal.

[28:30] 发言人 1: No, I can move from nowhere else. So this is not under consideration anymore. So I look at my fringe, the current goal under consideration at E3, E, E, and the P.

[28:43] 发言人 1: so this email in particular have depth of two so i will go to this. So after I go to E, all these I have already considered right. I can either go to H, I can go to R, I can go to E or P.

[28:59] 发言人 1: So. H and R have that equal to three, these two have that equal to one, so I definitely either be H or R. So according to our convention, always pick the one to the left.

[29:10] 发言人 1: So I go to H, so from H I can go to P and Q. I can still go to R from E, go to E from S, P from S. So that’s one, that’s three, that’s four.

[29:23] 发言人 1: These two are… That’s four according to my strategy. I need to choose these. So I choose the one on the left according to my connection.

[29:31] 发言人 1: I go to P, from P I can go to Q. This Q is beyond the consideration. This R is beyond the consideration.

[29:39] 发言人 1: This E and this P is beyond the consideration. These are yellow. Okay，so I picked the one was the deepest，I go to Q。

[29:51] 发言人 1: I check if this is go not go, so I stop here. So this part is no good, this part no no good the current. Notes that in my French under consideration in my French is Q, R, E, and P.

[30:07] 发言人 1: Out of these four. Q has the it is the that’s so I go to Q。Out of the queue, I do a goal test.

[30:16] 发言人 1: It’s not my goal, so I stop here. So all these are no longer valid. The current node that’s under consideration are R, E, and P.

[30:26] 发言人 1: Out of these three, R is the biggest. I move on to R. From R, I can go to F, go to E, or go to P.

[30:36] 发言人 1: Of the D three, as is the deepest. I go to F, perhaps I can go to C or G. I can still go to D or P.

[30:44] 发言人 1: Of these four, this two has the deepest depth. By convention, I will go to the left first. Okay, I need to choose them. At this point, I don’t know if I can take them.

[30:57] 发言人 1: So by convention I go to C, out of C I can go to A and I can still go to G, I can go to E and I can go to D. So out of these four, A has the deepest depth, so I will go to A. I do a gold test, that’s not why I want so stop here.

[31:13] 发言人 1: So the note on the consideration are G E and D. So these three, the gold G has largest to that, so I move on the G, I do a gold test this circle, so I find a solution with this. That first search algorithm is going to give me this particular route, and I have visited anywhere from here.

[31:39] 发言人 1: and this far. 哎，你把门锁了。Is this popular? How does algorithm work?

[31:49] 发言人 1: For example, in the midterm, I might ask you to step by step write out all these cues and all these phrases, all the notes on the concert. Okay, so I hope. Very simple, just remember visit the depths in this familiar bus.

[32:07] 发言人 1: Okay，in today’s tutorial，you are going to see any good information of this，or you want to try to implement this yourself。Okay，so any question? Okay，so if now we can take a look at the properties that we are interested，completeness of the normality，complexity space complexity。

[32:28] 发言人 1: Okay，so what are these four properties of our dead person? Okay, so now if you take a look at the top right figure。So the third time.

[32:41] 发言人 1: Of that first search is, let’s say we go with the convention that we use always visit if there’s a tie in the depths we always visit the leftmost note first then the search pattern look like something like this right you always search the leftmost. triangle that you keep moving like this way until you hit. They don’t know, okay?

[33:09] 发言人 1: So we basically have the search pattern of always visit some lab prefix of the tree that is under the convention. You always visit the left node first. You can choose to visit the rightmost node first if you want.

[33:22] 发言人 1: Then the search pattern will go from right to left. Okay，so in the end，if your goal happen to be the lower right corner。Then you may, in the worst case, you may ended up visiting the whole tree.

[33:41] 发言人 1: Okay，if the total depth is finite，it means that if you don’t have loops in your state spacecraft，then the time complexity is basically be raised to the power of n，because at worst case you may end up visiting one of them。And they totally have these many notes. So the time complexities be all in order be raised to the power of n.

[34:12] 发言人 1: Okay，so now is this algorithm going to give you a solution if the solution exists？Okay，the answer is yes，as long as you don’t have lose，because remember at worst case she’s going to visit the entire search tree，so if there’s a solution within your whole search tree，it’s going to give you，as long as the m is finite，because you will always visit the leftmost first，so if this m is not finite，then you will be trapped。LIKE THIS.

[34:48] 发言人 1: Without going to the rest of the trip, otherwise you will always find a solution. Okay。So how much? Space, what’s the space complexity? How many notes that you need to keep track? Basically, it is in the order of three times that because you anyway only looking at the child of a particular note and these many of notes remain in your frame is basically three times the total number that’s that.

[35:26] 发言人 1: Right, because as a worst case, you are going to the mass depth and you are having that many of the manufacturer. Okay，so from the time complexity perspective，it is pretty big，pretty bad，because you may end up populating all the tree，but in terms of space，it’s still okay。Right, so is this algorithm going to give you the optimal solution? By looking at this cartoon, I think the answer is obvious to you, right? This.

[36:03] 发言人 1: Solution is going to be the solution with the minimum cost in terms of the dance, number steps taken to reach it. But because you fix the command going on left, then you can never find. It will be the same as if you going on the right, if the solution exists like this, then you cannot find it.

[36:22] 发言人 1: So in general, no, the algorithm is not guaranteed to give you the optimum solution. Because you’re always going to either looking at the left or right prefix of tree. And you are going to imagine that before you see everything.

[36:39] 发言人 1: Okay，any questions on this part？Okay，I hope it’s intuitively now。So，um，let’s take a quick10 minutes break. We will resume back around10:30，then we will talk about breakfast search and uniforms. Okay。

[37:40] 发言人 1: 而且一万个全部都是打进去的。

[38:27] 发言人 1: 我们当时讲那个啥设计线的时候，说你从右边开走走走走走到走到头走这么回，回到下一个你能走的点儿，走走这回。

[40:21] 发言人 1: 所这看刚刚课前提了一句这周下周还会讲一下关于项目的就是关于目作业的但但但应该不是。不是这个什么或者这种。

[41:05] 发言人 1: 这次。

[41:16] 发言人 1: 这个角度还是对这个角度。

[41:26] 发言人 1: 还得在我把两个证据都放一块儿了然后中间删一些就是没有用的东西六十六他现在只讲了十几刚好三年三乘二十二。

[41:59] 发言人 1: 你说。

[42:31] 发言人 1: 他会。

[42:53] 发言人 1: 这个。

[44:06] 发言人 1: 四。

[46:50] 发言人 1: After creating temperature spray，let’s resume back。So，um，for the first，that first search，we kind of go very slow and show you step by step。But for the remaining.

[47:00] 发言人 1: into we will go slightly faster because they are exactly the same okay the only thing different is which one we will choose okay so now the next one we are looking at is called the graph. Research，okay，basically you want to explore the breath first before you going down all the way to the deep，okay? So now if you look at the surgery，now instead of the populating the deepest node first，you will always populate shallowest node，the node with the least。

[47:36] 发言人 1: Amount of depth, so at the beginning it becomes very simple, right? The same basically you have D, E and the P, they all have the same depth equal to one, so you just... they are all having the same depth, so you are just picking one out of some convention. Let’s say we keep the same convention of going to the right.

[47:55] 发言人 1: 然后你 publish B，然后 you can go to is either B C E。E or P, so out of these two set you have this equal to two and this equal to one. So this one tells us to choose the shallowest node, that means I can either choose E or P.

[48:13] 发言人 1: So we have the convention going to the leftmost first, so in here I’m going to choose E right? So after I go to E, I have B C E F H R and P under consideration. So out of these, the one with the shallowest node is the P.

[48:34] 发言人 1: So I visit the P, so after I visit the P, I basically have all these open, so I call with the one with the leftmost as my convention, I go to P, so I have these remaining. My connection will help me going here, right? So on and so forth.

[48:53] 发言人 1: Okay, so basically you get a picture, right? So the same as that for search, the we can differences you. Take it the leaves steps or the showers, okay?

[49:06] 发言人 1: So now you look at the search characteristic of grass for search using this cartoon. The thing you notice is that for a net for search we have said we will always visit. The leftmost first, right? And now, if you look at the breadth search, because very different.

[49:27] 发言人 1: He actually visits all the node. At a particular desk first. Before you move on to any further depth. So the cartoon grows like this.

[49:40] 发言人 1: Whatever how big the the surgery is, you will explore more on the node at a given depth before you explore any further depth. So it grows like this. Okay，so it grows like this way。

[49:57] 发言人 1: Okay，so what it actually means that you will process all the node above the shallowest solution。Okay，so let’s say the depth of the shallow solution is a variable s。And the maximum depth again, it is m.

[50:16] 发言人 1: Okay，let’s say the shallow solution occur at S，okay，so they are equals，then the search time equals to is basically be raised to the number of S, which is shallow to them，because it will process over the above the shallow solution，so anywhere above this S。You will always visit all the trees, all the notes, and you begin the discussion. Okay, so how much is the fringe going to capture? Again, it is the race to power tracks because you will actually put everything.

[50:55] 发言人 1: into the fringe because he needs to process everything. Okay，not like in the depth first search，which you only capture B times and it’s not in the power anymore。Okay，because you only keep a leftmost and you always visit that depth，there’s no solution，you don’t have to put that into page anymore，you move on the second time。When you hear, you actually capture everything, okay? so as you can see again if.

[51:32] 发言人 1: The shallowest depth of your solution is finite, and this solution yes, that it is guaranteed to give you the solution, and it is also guaranteed to give you the solution of the optimal solution if the optimal cost, if you have a constant cost for moving one to the other. Let’s say now, if you move between any node has a constant cost, then yes, the solution is. This algorithm is going to give you the optimal solution if you have a variable cost.

[52:06] 发言人 1: Let’s say you go for any node to the other node has a variable, has a changing cost. Okay，like in the traveling in Romania case。From one city to another city, the actual distance can be different.

[52:21] 发言人 1: Then this algorithm may not give you the optimal solution. But otherwise, if you assume any move you take, the cost is finite at the same, then it is guaranteed to give you the solution. Because he actually process all the solution before the shadows solution does.

[52:42] 发言人 1: OK, so now, as you can see from your mind, this the breadth first search is good in terms of optimality, because it will always give you the optimal solution, assuming that you have finite depth, assuming that every. Movement of one onto the other is constant because, but if that search algorithm is more efficient in terms of storage, it doesn’t have to put everything into the fringe. So the next question comes into mind: Can I have the optimality of grass search at the same time having the space complexity efficiency? From the desk, how can we combine them both together?

[53:30] 发言人 1: Okay, so this will debates the second and the third algorithm called iterated deepening. that’s first okay so what they want to do like i described is to get the benefit the space the space complexity benefits of that first search and the same time with the shallow solution the optimal solution advantage of fracture search. Okay，so what he does is he actually wrong。Say bfs，the depth first search algorithm，and variable depth limit。Let’s say I want bfs at depth limit equal to one，if no solution，I increase the steps limit from one to two。And run it again, then I run from two to three if no solution until I increase the steps to enough such that I enter.

[54:27] 发言人 1: I found a solution. Okay，so if you think about this a little bit deeper，what you actually did is message。This is the surgery.

[54:44] 发言人 1: So what does iterate the big big algorithm does is that it gives some limit. Okay，after you adding this limit，your DFS will always explore this part，thus no solution you move on to the other。So this red line on this definitely prevent you go for all the way to the limit, limit over here.

[55:07] 发言人 1: You always going to this red limit you sat first. Then you move like this. So you can imagine this works.

[55:16] 发言人 1: 嗯，哎，就比例一点。Of this depth limited DFS works like a DFS, right? Because he actually explores all the solutions before you move on to any new depths.

[55:31] 发言人 1: But at every time, he always explores the left-most prefix of it, except for processing everything as you. Okay，so one thing，if you there there is no solution，you lose the steps，increase the steps to another depth，then you again。对的。

[55:56] 发言人 1: okay，if no solution，increase this again。you do this. Yeah, okay, so something immediately you will notice is that what happens to this. Let’s say I have a solution in here, I found a solution in here, but. You were wounded, oh!

[56:19] 发言人 1: All these part I repeat many, many times because that equal to one I visit the whole part, that equal to two I visit this whole part again, that equal to three I visit all these parts again. Right，so how much is this space compared to the savings I have from the that person? Is.

[56:45] 发言人 1: This question try to answer. Okay, so if you actually think about this, remember the branching factor is big. Let’s say the shallowest depth is s.

[57:00] 发言人 1: So the most compute you are going to spend, or if you look at the big old complexity. Or the space complexity in this case. It is actually in the order.

[57:18] 发言人 1: Was the matching that to the house? So if you put this in calculation, at the first step given the branch factor of three, you weigh three. If that’s equal to two, he weighs let’s say four, he weighs sixty.

[57:38] 发言人 1: And third, that’s you would raise40. So actually, as you go down, most of these computation is actually happening on the last layer. So the computation associated with just the last part of it is way more than everything combined before.

[57:59] 发言人 1: ok. So the answer to this question is yes, in general there will be some waste, but this waste is not governing factor because most of the compute spend is anyway at the most steps. Because that’s where you have most of the notes.

[58:18] 发言人 1: Okay，so remember when you calculate the notes，you have this one plus B plus B square plus B cube all the way plus BM，right? So everything before this last level is considerably much smaller than this V S. So the order is in the V S.

[58:37] 发言人 1: If you you can put in the number or you can multiplicatively write this. It’s very simple. Okay，so this part is something。that we do repetitive work, but fortunately most of the work is happening here, so it’s actually okay or at least manageable. okay, any question on this part?

[59:03] 发言人 1: Okay，so now，so far，we have been talking about search algorithm，assuming the cost going from one node to the other remains as constant, but in reality，like the traveling in Romania or in any real world cases，that’s

usually not。Very realistic to assume. So now, from now on, we will look at another set of algorithm has been created for cost sensitive search.

[59:31] 发言人 1: So what it means is basically because for you to move on from one state to another would not be constant anymore. For example, from S to B, you have a cost of three for S to B, you have cost of nine for S to B. Okay, so the previous two, including the iterated division, the previous three algorithm will give you the shortest path in terms of numbers of actions。But no least number of action do not necessarily means the least number of cost.

[60:08] 发言人 1: If you think about traveling in Romania case, it means you will visit the least number of cities. Bahrain to Bucharest, but minimum number of city does not means the minimum traveling distance。Okay, let's say if you walk from here to Beijing。Suppose you are flying all the way to UK and make connection to Beijing.

[60:36] 发言人 1: That's one round. You only stop one city in UK. The London for example Beijing.

[60:44] 发言人 1: You only visit one city in between, but it's much, much considerably longer than if you go from here to Changsha, Wuhan, Shanghai and Beijing. Right? Even though you visit a lot.

[61:00] 发言人 1: City but the total travel distance is way less. They're going round around UK, for example. That's what we are talking about here. Okay, if you know the cost, probably between any two states, then we have a new algorithm to solve it. Okay, so the first one we are going to show you is called uniform call search.

[61:24] 发言人 1: Okay, so what he does, very simple, you will always move on to the next node with the minimum distance away from you。Okay, so take this as for example, if I start with S, I can either go to D, E or P。Right, if I go to D, the traveling cost is three, if I go to E, the traveling cost is nine, if I go to T, the traveling cost is one.

[61:56] 发言人 1: So this uniform cost search algorithm will always tell. Choose the one with the cheapest, the least cost way for me. So in this case, I'm going to be.

[62:08] 发言人 1: Right because that's cheapest from P I can either go to Q or I can go to E or I can go to D. Right, so if you take a look at the skill. the number we show over here.

[62:28] 发言人 1: Remember, any node in your search tree is the path trace all the way to back. So the number we showing here is not the cost from you to go from P to Q. It is actually from you to go from Q all the way to X.

[62:48] 发言人 1: So if from S to P is one, from P to Q is fifteen, then the total cost for my Q is actually sixty, because every node in your surgery is a path. Trades all the way back to the room. Okay, so now the three most under consideration is Q, E, and B。The cost will tells me B is the cheapest one, so my view is the uniform cost search is going to move on to D.

[63:17] 发言人 1: From D I can go to B, C and E again. Remember this number showing over here is the accumulated cost from this node all the way to root, not just to its parents. So for B, you see over here from B to D is only one but D to S is three.

[63:36] 发言人 1: So the total number of B is actually four okay? So the one answer your consideration when our fringe is B C E E and Q which will cost of four eleven five nine and sixteen. So out of these, obviously B is the choice.

[63:57] 发言人 1: So I’m going to move on to B. With B, I can go. A C D E Q are still under consideration comparing all these the cheapest cause is E has caused by.

[64:11] 发言人 1: So I’m going to move on by. Okay，so from E I can go to H or R and C A E Q still on the consideration。The amount of them, let me see, the one of the cheapest cost I think is eight. So the algorithm will ask us to go for it.

[64:36] 发言人 1: Now the goals of this process can be on one under consideration C H R E and Q out of these R seems the one that’s the cheapest cost so I need to move on to R for R I can go to F. And I still have H C E and Q under consideration. So how these let me see.

[65:06] 发言人 1: It’s probably this app. But it’s achieving because of eight. So I move on to this app.

[65:13] 发言人 1: For app, I can go to C, G, I still have H of C here, E here and Q here. So the minimum costs now becomes E. So this search algorithm is going to ask me to move on the E.

[65:29] 发言人 1: I have H, R, Q, C, G, H and C under consideration. So all of these, let me see. If my map is correct, I think G has the lowest cost path.

[65:46] 发言人 1: So I move on to G and do a protest. This G, so I’m done. So I pop up one interview. Okay, so this is my solution.

[66:00] 发言人 1: So something to remember for the uniform cost search in applied problem where moving from one state to another state incurred a different costs, and it will always choose a state which are closest to me first. So you will always expand know. 我是去年思考。Okay，any question so far? It’s very similar，only thing changes against the strategy。

[66:33] 发言人 1: Okay, nothing change. So now, if you think about from this again cartoon kind of perspective, then how does this UCS cause actually explore this tree? Explore this tree in a non-uniform way, right?

[66:51] 发言人 1: Explore the tree in this kind of uneven polygon shape, and this shape is really determined by the call. Right, if you still remember when we going over the street, then this is the. 拍的。

[67:11] 发言人 1: landscape we are visiting right this is really uneven because all these are happening with different costs et cetera so and this exact shape depends on the the actual cost so you actually explore the the treatment so in summary what it does is that it will process all the node with the cost that’s less than the cheapest solution from the starting point. Before you move on with any solutions, this higher cost that’s my definition will happen. Okay，so now let’s do some assumptions。Let’s say the actual solution causes C star。And every at our cost, so on average the cost from one node to the other is at least.

[68:00] 发言人 1: Absolutely，then the effective depth，this on average the expectation of the or the in some sense the average depth，you can calculate roughly C storm divided by F。Okay，so this is where the solution happens。And the total time defining this factor is basically defining the successful steps is basically useful for us to calculate the time complexity and space complexity.

[68:33] 发言人 1: Okay，so the time complexity is basically the similar to the breath first or depth first search，the only thing different is the depth，where in the depth first search it is the max and the maximum depth，in the breath first search is the shallow solution depth，in the uniform cost search is the effective depth。Okay，everything else is the same。And the French is similar to Brax’s research because.

[69:05] 发言人 1: We actually consider all the solutions with the cheapest cost first. If this chips of cost determine the expected depth, then we actually consider all the points prior to that. So you actually need to put everything into the frame.

[69:20] 发言人 1: was cost less than. We currently consider the lowest cost. Okay, so if you go back to this tree following our kind of derivation, you can see that it actually covers almost all the notes prior to a particular depth.

[69:39] 发言人 1: So it’s very similar to to breath verse okay，and of course this is complete because he actually covers everything。Okay，so what does actually uniform code search does? If you think about his.

[69:58] 发言人 1: Um，actual search behavior，this kind of。Uniform kind of a polygon，the one very big issues。About this search is that he searched with the center around himself, the starting point, which is where is the actual destination.

[70:23] 发言人 1: He has nowhere in fact into the. Starting one, okay? So if you’re thinking about this call to life of how you propagate starting from the spark to the stop in your mind, if you know the search is going to like this.

[70:45] 发言人 1: And in an ideal situation, what you want is I can directly go from start to stop. That is the faster route to find it, right? But in the unit.

[71:00] 发言人 1: For course, what actually happens is the most remember it will process all the solutions to the cheapest cost to the starting point before moving on to any other. increase the cost of units. 因为我们的 C.

[71:19] 发言人 1: This blue circle denotes all the solutions with a constant cost from my starting point. Then you will visit all these before extend the. circle again. Then I will visit all of these before I expand the circle again.

[71:36] 发言人 1: Okay，even though the most efficient direction to my destination is going this way，but you will always like moving this way，and you find this。Okay，so conceptually，all these parts。I wasted because it’s only completed the other side of your stomach.

[72:05] 发言人 1: So is there anyway we can improve this? Okay, this is what we are going to cover next. Okay, so if you think about the schematic, this is basically just a small cartoon showing you that if you run this in the concourse search, you will expand all the directions before you move on to next.

[72:27] 发言人 1: Okay, so before we show you how we can solve this problem again, I want to bring up this issue which we have been bringing up almost every single time we’re talking about it, including I think twice in the previous lecture is. We never search on the direct on the actual problem. We always derive a model of the problem.

[72:51] 发言人 1: Ok, so if you thinking about what we mean by the model is this readers. overlap is model of the actual let’s say b could be a particular city as this is this as someone you can say. And this goal can be Beijing.

[73:13] 发言人 1: Okay，so we have the exact right on all these，but we never solve the actual problem。What we do is we drive it out。Let’s say in the actual world, Guangdong，maybe Guangzhou。

[73:26] 发言人 1: THE ACTUAL DISTANCE IS MAYBE A HUNDRED KILOMETER BY CAR, BUT TO MAKE PROBLEM SIMPLE WE SAY THIS IS JUST THREE. And all we want to keep is let’s say from Guangzhou to Changsha is eight hundred kilometers. Maybe people three will be better, so all I’m putting here is the cost is relative proportionally correct to the actual.

[74:01] 发言人 1: But we never use the actual one here. Because in the after world things can vary even the distance from 东莞到长沙，only 东莞到广州，only100 kilometers，but depending on the traffic，you may have the car broken down the road，so you might have to take the bus one。okay，same distance，the actual traveling time also varies a lot。

[74:25] 发言人 1: Okay，so all these things actually，all the real world complications are there，but when you’re actually thinking about and trying to solve，you do not consider all these real world practicalities into designing element，you assume all these are in the perfect world in some sense，and you use them all to do it。Okay, so in the end you need to be really careful because how you will do really depends on your model. Okay, nowadays it’s pretty good.

[74:56] 发言人 1: Okay, all this model we have, but here I’m showing you. you know ten or twenty years, ten at least fifteen years ago on the google map where if you go from this city to this city. If you have a bad model, you actually take you around like this.

[75:14] 发言人 1: Okay, so your search algorithm one to show you is your search algorithm will work as long as you have a good enough model to approximate what you actually want to solve. Okay，so any questions so far before we move on to more slightly better algorithms that this beautiful search algorithm for the cost。Any quick questions?

[75:48] 发言人 1: Okay，so if now we will move on to second part of today’s lecture where we talk about inform the search matter，where we will introduce the concept of heuristic。Which is actually very important and useful now, even in nowadays for large language model era as well.

[76:06] 发言人 1: So on top of this, we will talk about two particular algorithms that build on top of this concept, which is called greedy search, which you must have used before, and a start algorithm. Okay，finally we will talk about how you can

obtain the existing answer. Okay，so in the first part of today’s lecture，we are pretty much on this side.

[76:32] 发言人 1: All inform the search from solving the problem perspective. Basically, they imagine this to be search with your eyes closed. You never use problem information into your search.

[76:45] 发言人 1: It does not really care what you want to serve. I always go by this rule, like I always with the deepest snow first, shallow snow first. It never, it does not really care about what the actual problem is.

[76:59] 发言人 1: Okay. Where the informed search based readers is how can I incorporate from the information into the search itself, and this problem information is being captured by this concept on a heuristic. And we can design either a radio or even more fancy algorithm to utilize all these.

[77:19] 发言人 1: Okay，so if you are. If we are thinking about this problem again, coming from this textbook, what we want is this: imagine this to be our ghost day. We want the pancake to be arranged from smallest to the biggest.

[77:43] 发言人 1: Okay，now all these configuration you see over here are not in the order like this，all the smallest to the biggest。Right, so now your search algorithm wants to divide the way. Putting them into the food into this particular road state.

[78:05] 发言人 1: Okay, now we want to measure how far away of each one of these. To this and the way we do this is maybe through a concept called heuristics. Okay, so let me introduce the heuristics first, then I’ll show you exactly how it does.

[78:23] 发言人 1: Okay, so mathematically. Or conceptually, what a heuristic means is a function. That ask me how cause a say to be so。

[78:37] 发言人 1: Let’s say we pick this pack that case. This is you see over here. is a state.

[78:49] 发言人 1: You want to ask me how causes to a goal. Let’s say in here the goal is for this pacman to eat this ball. So if the state becomes the terrain then.

[79:00] 发言人 1: Here without the ball, any of us. So now you want to ask me how far away is basically this particular state to this ghost state? Can be approximated as how far away your packman to the stop.

[79:18] 发言人 1: Right, so what you can do for this particular case is you can estimate. Maybe the Euclidean distance could be a heuristic function for this problem. Okay, so what I’m trying to tell you is a heuristic is a estimation function that tells you how close a particular state is to your goal.

[79:46] 发言人 1: And unfortunately, it has to be designed for any search problem you want to solve. There’s no. And economical heuristic, meaning that you cannot do it once and hope you will work for all.

[80:01] 发言人 1: No, because that varies from problem to problem. In this particular case, if you want to estimate how far away the state is to the finished state of Pacman eating the door, you can use the Manhattan distance. Basically, x plus y is a

Manhattan distance, or you can use an Euclidean distance, which is basically x square plus y square over square.

[80:28] 发言人 1: Okay，so those two can be your tourist function to ask。Okay，any questions so far? Okay，so now in this. Um，traveling in Romania is we can actually give you some touristic，so remember in this original problem，our goal is to start from Iran and we want to write as Bucharest。And every brown showing here the number.

[81:00] 发言人 1: Right next to the arrow, the the the arc is basically the actual traveling distance from Iraq to this supreme city. Now in here, we are giving you this. H heuristic，I use function H variation，its input is a C。Okay，so if you input a ride here，I will this number366，what tells you this around the city has an estimated distance away to my Bucharest in front of366。And it is a straight line distance.

[81:47] 发言人 1: Okay，and we are using this as a units。Okay，now for this pancake case，what we can use as a heuristic is the index。of the largest.

[82:00] 发言人 1: Cake that is still out of place, so let’s say this is our target: the smallest on top, the largest on the bottom. What this function tells me is where which is the largest cake that’s currently out. So if you look at this one.

[82:18] 发言人 1: First of all, if you look at this, let’s say I gave it index: the smallest tanking index one, the second smallest index two, the third index three, the last one index four. The index4 pancake is in the right order, and among the other three, none of those are in the right order, and the largest of them is three, the index number three. So I can use the street here, it’s the number three for it.

[82:46] 发言人 1: In this part, we can see the index4 pancake is also out of order, so the here is six four. So on, so forth for here also the index four and it is on board. ok.

[83:00] 发言人 1: Again，the heuristic function is a function that we design as human engineer to tell how close or how far away a particular state to my goal. This may not be actually accurate，it is just an estimation. If you design this well, it will works perfect. Otherwise, you may not work.

[83:23] 发言人 1: Okay，so based on this heuristic concept，the first algorithm that people develop is called3D search，and it is very very simple。It is the same as previous uniform search algorithm，the only thing changes is instead of going to the cheapest node first，it will go with the heuristic function，whoever talks，which whoever node has the closest。THIS IS COOLEST.

[83:54] 发言人 1: to be gold，i want to reach，then i will be. So for example, from Iraq I can either go to Tehran or this team city or this Iran. Now if I’m starting from Iraq, I can go to, let me see.

[84:13] 发言人 1: I can go to Seville, which is right here. I can go to this team city, which is right here. Or I can go to the Iran city right here, and this number is my heuristic.

[84:32] 发言人 1: Now what this greedy search tells us is, I simply pick the one with the smallest security number because that tells me that is why I ask me to be the coolest to my goal. So in this case, I will go to this view. Okay，after I go to school，I can

either go back to Iraq or I can go to Bulgaria，I can go to R city or this Arabia，okay, so。Regardless, I have178.

[85:03] 发言人 1: I go to Iraq, it’s366. If I go to RCD,193. If I go to Iraq, that’s380.

[85:15] 发言人 1: So out of all these, the smallest is for guards, right? So my video algorithm is going to help me to go to Budapest. Budapest, I can either go to. 四十五 or I can go to Bucharest. So if I compare these two Bucharest has an estimate of zero whereas two has estimate of three hundred and fifty three my algorithm tells me to choose the smallest. Age function, so I go to Google.

[85:47] 发言人 1: Okay，so it is just the same as uniform concert，where the cost function has been changed from the actual distance that you need to travel to the next city to a heuristic。Function which estimate how far, how close it is from this state to the goal. So we change from the actual travel distance away from me to the expected travel distance to the goal, which is an estimation.

[86:19] 发言人 1: Any question? Okay，so now after this in your mind，what do you think? What can go wrong? Okay，I think，I hope this is some more clear to you。The only place or the actual main concern for this algorithm is this.

[86:49] 发言人 1: how accurate. はい、そうです。If your heuristic is completely correct, Then no worries, you will just give you the good.

[87:00] 发言人 1: SolutionTION, if your risk is actually wrong, becauseSE in the end it is just some asc. you ask to me how far away the state go. you have not actually triedED.

[87:11] 发言人 1: you may not actually know. this is just some asset. okay? SO if you think about greedy surgery, it is actually expanding a note that you think.

[87:21] 发言人 1: IT’S COOL. 不能够说，by the heuristic function。Okay，in the common case，it will just strictly take you to the goal if your heuristic function is informed and correct。

[87:41] 发言人 1: If your heuristic is completely wrong，let’s say the heuristic for your traveling distance is the actual distance。In the reverse order, then you will simply take you to a completely different, completely wrong code. OK, but in the end.

[88:00] 发言人 1: You will reach your goal because it runs just like that’s for a search. You will listen all the notes, even if the heuristic function gives you wrong, but in the end you will explore the whole tree. Okay, so the efficiency of the screen research is actually determined by this heuristic function.

[88:20] 发言人 1: Okay，so now for this case，if my heuristic function is the actual distance from the screen to the red，instead of going this way as uniform search，if I gave an actual distance from the screen to this red as my heuristic。What you will happen is you will raise no time on the right to go there because my heuristic function is aligned with the actual distance. Okay, so again, if your heuristic is correct, then the video algorithm will give you.

[89:00] 发言人 1: Correct answer, but if it doesn’t, if you have, if you can expect some error, then it’s so get p, and that is the whole motivation for A star L because in

reality. You may not always be able to design a accurate and good function. So in the end, a star make a balance between the uniform call search and a really good and a really search.

[89:27] 发言人 1: Okay, so the whole idea, if you’re thinking about a uniform call search again. It is basically a very conservative. You always visit.

[89:41] 发言人 1: the node that is closest to be first, because that is the minimum investment i need to make in some sense。Right, if even if I choose the wrong one. But that note is the already the lowest cost that you respect.

[90:01] 发言人 1: So if it is they have to be made by uniform call search, that will give me the minimum loss because that is the already the minimum I need to spend correct? Even if that’s wrong, the cost is still better. That is the mindset behind uniform call search.

[90:20] 发言人 1: Okay, so if you thinking about this particular. This particular search problem very simple, the uniform search algorithm will basically explore the search as follows。Okay, the goal is over here。And if you apply the uniform cost search, you will just look at what is the cheapest cost that you can travel.

[90:46] 发言人 1: So here you will go to A, for A you can go to B, C and D. Right, so if you go to B, the total cost is two；if you go to D, the total cost is nine；if you go to D, the total cost is four. So you will definitely go to B first.

[91:02] 发言人 1: From B, you can go to C, the total cost is three. If you go to D, the total cost is four. If you go to E, the total cost is nine.

[91:11] 发言人 1: So you will go to C. After you go to C, you will go to D. Right? The total cost from here is four. This is nine. So you will go to D, then you go to G. No, you will go to E actually, because go to E, no, the direction does not go to the place, so you will go to G.

[91:29] 发言人 1: Okay, this is what happened with uniform search。Now if you go to greedy, what this greedy tells you is you need to go by this heuristic number。Sorry, this is not very visible, but if you see this H function right next to every node is the heuristic。

[91:51] 发言人 1: So from that you can only go to A, right? From A you can go to E, B and A. So now if you take from B.

[92:00] 发言人 1: Heuristic number is six from B, the heuristic number is two from E, the heuristic number is one. So out of B, D and E, you will go to E because the heuristic number is one and very low. From E, you can go to D and the B is still under consideration.

[92:22] 发言人 1: Right, from C the heuristic is two, from B the heuristic is six, so you will go to B, then you will go to C. Okay, so now what does uniform, I mean what does A star algorithm try to do is. Instead of going directly by the cost, so this number here is geo, okay?

[92:50] 发言人 1: Instead of directly going by the actual traveling distance or completed by this estimation of how far away the state is to the goal. Can I combine

them? I also consider how far away I need to travel, the actual investment I need to do together with the expectation if I travel there what’s the future cost I need to do in order to reach the goal.

[93:20] 发言人 1: So this is the future expectation and this is the actual investment you need to do at currently. So these are algorithm make a balance of these two by combining them together. Okay, so mathematically very simple, you just combine the distance you are going to travel for a beautiful global search plus the heuristic function for the estimate with the future away to the goal.

[93:45] 发言人 1: Okay. So what it does is if you actually implement this, which you will do for today’s, for this week’s. Uh, tutorial, you will see more。But one thing I want to point to you in the classes, if you think about the optimality of the Easter algorithm because the heuristic is really... I mean, the intuition is very good right?

[94:12] 发言人 1: You make a good balance between what you immediately need to investigate versus what’s the expectation you need to investigate in the future. So it sounds very nice. But if you look at.

[94:25] 发言人 1: THE ACTUAL PERFORMANCE OF A STAR, YOU WILL NOTICE SOMETHING. SO FOR EXAMPLE IN THIS CASE, LET’S SAY VERY SIMPLE, THE GOAL IS FOR YOU TO GO FROM S TO G. Okay, so now if I go by uniform.

[94:46] 发言人 1: What he asked me to do is either A or G, I should go A because that has the slower distance by S way, so I go this way. Then from A to G, and already G, I can choose I go this way, so I ride the G. Now if you apply A* algorithm to solve this, what you will have is remember the f of n equal to g of n plus h of n.

[95:21] 发言人 1: Now if you compare this node versus this node. the note for going to. A node is two. Plus is harder he will succeed.

[95:42] 发言人 1: If you now calculate machine, it is the actual traveling distance of five, but you write it to zero. So it is fine. Okay, so what your A star algorithm going to tell you to do is go with。

[96:02] 发言人 1: Directly to G, so I go this way. But if you take close a look, if you go around a, the actual traveling distance. is lower than directly going from S G.

[96:20] 发言人 1: So this eight star algorithm is not guaranteed to return you the optimal solution. Even in this very simple case, but you may wonder. The A star algorithm design looks very nice, but in the end why in the stock return the optical solution.

[96:43] 发言人 1: Again, it happens in this age. 嗯。The heuristic function you provide, it is not an optimistic estimation, instead, it.

[97:00] 发言人 1: Your estimated cost is more than the actual cost. Then A* algorithm is not going to guarantee to give you the optimal solution. It’s not guaranteed to give you the cost, the path with the least number of cost, least cost.

[97:18] 发言人 1: Okay, again, if your H function, your ASMA function is not optimistic or in some sense later you will see it’s impossible. So if your heuristic function. The

estimate is not less than the actual cost, then there is no guarantee for that base are algorithm for optimality as you see over here.

[97:41] 发言人 1: The actual costs two but a but heuristic estimate that is six but a to g the actual costs two. But he asked me to be six, so your external algorithm is asking you to do so this round. Which is more expensive than for eight for s to eight eight g which is four.

[98:08] 发言人 1: 啊，可以，你能扣的分儿。ok，so。In the end, you will see for all these algorithms to work, all you have to do is design a optimistic heuristic function. Okay? So more formally, this kind of heuristic function is defined to call a disposable or this concept is called admissibility.

[98:35] 发言人 1: Okay，yet an inadmissible heuristic is from the mindset being pessimistic，being really critical thinking，you always want to be on the safe side, you always want to overestimate things。You’ll always want to be prepared for the worst case scenario. Okay，so in a means you always thinking from the worst case scenario，you want to overestimate so be on the safe side。And optimism means that you are a happy guy and you are very optimistic.

[99:14] 发言人 1: You always think that things will happen in a good way, so you never want to prepare for the worst case. You want to be optimistic, means that you are very optimistic. 你相信事情不会那么糟糕，所以你要 optimistic。

[99:33] 发言人 1: ok，so you think the actual cause is always going to be smaller, right? it’s not going to be that bad. so a star algorithm actually require you to have a undeniable insight.

[99:47] 发言人 1: In order for it to actually go, and we call that an optimistic. So now, rather than optimistic, what? And the next for heuristic actually means mathematically is false.

[100:05] 发言人 1: Okay, so let’s say. H star is the actual cost. So remember heuristic is an estimate, and H star be known by H.

[100:19] 发言人 1: Now H star is the actual, so the estimation become perfect is my H. So you want to design a H in between zero and the actual cost. So again，it is the optimistic estimate.

[100:37] 发言人 1: So now，let me give you two examples. If you look at the packman. Previously we talk about if we want to estimate, if we want to design a heuristic for this type of case, we can either use Manhattan distance.

[100:58] 发言人 1: Which is x plus y as my heuristic. Or we can use this Euclidean distance. let’s call it h two which is. So if you, if we go with either H one or H two, and you think it. there or there divisible or not what do you think.

[101:22] 发言人 1: Are they being optimistic? Are they optimistic? Remember the definition is the admissible means in function you return the actual value is less than the estimated value is less than the actual value.

[101:40] 发言人 1: Do you think these two case are admissible or not? So let’ s take a look at it. So now, if I’ m in the states, what is the actual cost for me to go in here? The Pacman needs to maybe travel this way.

[102:00] 发言人 1: Way down this way, down this way, down this way, down this way. So the green curve is the actual cost for this tagman to reach this goal compared to the green with the blue. Is the blue less than the green or more than the green? It’ s actually less right? So this is actually invisible.

[102:23] 发言人 1: So what this characteristic function means is as if your pack that can go through all. It’ s my bad business. what does this. Euclidean distance means it means as if your partner has a plane, you can fly like a Superman. That is the Euclidean distance.

[102:49] 发言人 1: So both of these are acceptable because the actual distance returns by these two functions are actually less than the actual cost which you need to go around the obstacles. Okay, and the same thing will apply for the packet case. So I leave this as a exercise for you, then you can think about whether you see the largest index as an acceptable heuristic one.

[103:18] 发言人 1: Okay, so again, if now there’ s all these, if you’ re thinking about the optimality of the expert algorithm, you can actually go with the proof we show over here, it’ s actually very straightforward, so if you’ re interested, you can go over the proof and yourself, if you have partial company, and it’ s straightforward, it basically shows you if。You have a heuristic, you have a decent heuristic HM, then you can in the end prove that your algorithm is actually optimal. Okay, so now in the class, what I want to give you is a more holistic view, and also how you can develop a heuristic.

[104:02] 发言人 1: Okay, so if you now compare the uniform call search versus the A star search, what you will see is. Let’ s say your goal is over here. Uniform call search will always go to the node that is closest to me first, so we explore all the source space in this uniform.

[104:28] 发言人 1: kind of have. Right, then the greedy search will just go with wherever your heuristic want. If your heuristic is right, then it will go straightly.

[104:41] 发言人 1: And what the A star algorithm does is it try to scale this. Complete the circle into a little soil where you hope the long axis is pointing towards the goal. okay so.

[105:00] 发言人 1: If you gave in a heuristic which estimate how far away any node to the scope, you will basically skill this perfect circle towards the goal. If your heuristic is correct and again at least admissible, then you will be on. Okay, so this are basically two different kind of search mindset and search outcome in terms of content what will happen.

[105:29] 发言人 1: Okay, so now if you thinking about on a animation perspective, this is what happens. So you are not exploring. Everything like this before you reach, also you are not searching like a green vehicle directly like this because you assume this heuristic might be wrong.

[105:52] 发言人 1: You all you have like this kind of skill. Toward this open issue, so if you explore more. Or what you believe to be good, okay? But you are not completely trusting that because you think that may be wrong.

[106:10] 发言人 1: OK，so because if you thinking about in this. Your heuristic are actually perfect. So your elite soil is actually skills directly towards the shape. But if you think, if you think about this in reality, what may happen is you may have a bad age.

[106:37] 发言人 1: 然后你 looks like may may have to be skill like this。Right，and if you are running eight star，you can still keep this。But if you have a wrong heuristic and you are completely going with greedy search, then you go all the way like this.

[107:01] 发言人 1: Right so eight start is something that we want to be efficient at the same time we do not want to be over confident that wherever we want to get the end is correct. Okay，so we have this combination。Okay, so this is basically why I want to summarize the A star algorithm.

[107:25] 发言人 1: You can imagine this is a balance between the actual cost versus the estimation of the future cost. Okay，so here we call the backwards cost is something they actually need to spend versus the forward something they estimate from your university。Okay，and A star algorithm is actually guaranteed to be optimum if you have a decent heuristic，and again heuristic design is the key。And A star algorithm is actually been used in many many different domains.

[107:58] 发言人 1: Ok, like I said, if you are using ChatGPT, they also use A star. But they call it big search by this pretty much the same. Okay，so the last point I want to make for today’s lecture is.

[108:11] 发言人 1: Let me see. So instead of taking another break, maybe my plan is to quickly go over this and show you the tutorial. So maybe you can go for lunch slightly earlier today.

[108:25] 发言人 1: Okay, so I’ll skip the break and keep going. Can we go? Let you go earlier.

[108:29] 发言人 1: Okay, hope that’s okay. So, as you can see so far, A star algorithm works well, right? On one condition is that you have a way to designing a good hearing system.

[108:43] 发言人 1: Okay, so later on you will see particularly in the domain of deep learning. Many many of these efforts are actually going into how you can just define all these heroes, and for example in the deep reinforcement. Learning they have a big neural number to be a heuristic function.

[109:06] 发言人 1: Right, if you’re thinking about Alphago. Then later on in the two slides later we will actually explain this to you in more detail. Basically the heuristic function they use is a new method to estimate how good how that is.

[109:21] 发言人 1: Okay. And as you can see more and more such efforts going into designing a good heuristic function then people start thinking about can I use an AI? the design of the heuristic function.

[109:36] 发言人 1: Okay，so what I’m showing you here is by the end of2023，almost three years ago，Google DeepMind runs a AI，a large language model to design a heuristic function automatically。I give a problem，can you just design a heuristic function that works? Okay，so what the problem they first tried is called the online.

[110:01] 发言人 1: Okay，so what does online video packing process is you can see here，there’s many many things。Imagine this needs to be some containers, like the logistic company, they have many many cargoes where they can store things. Okay, so what this problem does is that you have some incoming items that you want to put into one of these bins.

[110:30] 发言人 1: Okay，so the restriction is if your beam has enough space to hold this item，they can put it，otherwise you need to put into a new one。Let’s say this thread can operate here，so you can either put here or here，so here you can operate。Okay，and what this online means is you do not know what’s the size and what’s the actual package is going to come come in，so how you don’t know after this red。What’s the next one? What’s the size of the next one? Just like in the realistic logistic kind of having the receiving scenario.

[111:08] 发言人 1: You don’t know what’s the customer going to give you as the next package. So this search problem wants to find a. packing strategy such that I can pack all the item using minimum number of base.

[111:26] 发言人 1: Okay，so this is basically the problem defined now。What it does is they express the heuristic。As the Python code imagine that could be some function or some algorithm some control being implemented in Python.

[111:45] 发言人 1: Okay，then he asked a large language model，back then they use a star coder，okay，something they develop，and they ask this language model to. Look at this curious function, can you help me design a bad one? So, a large language model gave three or four, I think in their case they tried five or eight different variables.

[112:17] 发言人 1: For each one of these heuristic function，you can put that into an actual simulation environment and check how good or how bad this heuristic is. For example, what they do is they can plug this into an A* algorithm as your H function, and then they run this online big problem. In the end, you have some score.

[112:42] 发言人 1: If the score is good, I put this heuristic into the database. If the score is bad, I throw the tourist away. then i got into a loop.

[112:55] 发言人 1: Where in every loop, every iteration, I sample. A heuristic function for my database, which has been found previously to be good. I ask my large language model to make some variations of this. I tried it out, put back into the database, otherwise I run this in loop. In the end, hopefully I find something works good.

[113:22] 发言人 1: Okay，so in here，they compare，they found the heuristic versus the heuristic that people has been used。So this one is what people or the logistic company like Sun Feng has been using. So what does functions basically says is, I will put the package.

[113:44] 发言人 1: into the being. With the smallest remaining capacity, but just enough to fit my iPad. I do not want to put the package into a completely open thing.

[113:59] 发言人 1: I want to. into being that was least amount of capacity left but just enough to fit my. Okay， and then on the right you see what large language model have found。

[114:16] 发言人 1: I don’t know what it does, but it does some calculation. In the end, it works better. Let me show you the actual performance of a lot is what human has designed how they will have exactly the same package.

[114:32] 发言人 1: In the end, what you will see is according to the strategy found by large language model, they use actually one less beam over here. Okay， and the tricks happens in the second。好。

[114:52] 发言人 1: In actually in the third green box, if according to the strategy of human have design, you want to put the box into the being. With the smallest remaining capacity but enough to hold it. So if you compare the B number one and the B number two.

[115:13] 发言人 1: A one has less capacity than B two but also enough to put my dream so they put the dream at the bus. But on the right, instead of putting on the first they put in the second without knowing that there will be a new red box. But after you doing this.

[115:34] 发言人 1: The third one, which you have no idea, comes with actually smiling bigger size. 那 actually can fit into the books。So in the end, they were able to save one thing.

[115:48] 发言人 1: Okay, so we all these things are happening to you. Maybe look at a little bit naturally, but in the end what they actually found is... So one, can we talk? What I want to show you is.

[116:00] 发言人 1: Heuristic function has been explained in the class has to be designed case by case for example in the pacman is the manhattan distance. In the pancake is. It is the largest index that’s out of order in this packing case， the online impact case， it is the rich being you should put your box into。

[116:26] 发言人 1: Okay， so all these are really problem specific， so unfortunately human engineer and human practitioner has to make these choice for every problem they want to apply a star。So you become something that is really manually have for any product you want to apply a start you want to develop this h function yourself otherwise there is no guarantee that your a start algorithm is going to work now into this ai stage where large family model can write code itself. Researchers has formulate the design of an algorithm, a computer science algorithm, a search algorithm, or any algorithm as a.

[117:12] 发言人 1: Search problem by so they formulated design of an algorithm as another search problem by letting AI to automatically search. So what it is is. They gave the problem description to AI and let AI sample many, many candidate algorithms, and then they provide an evaluation for it to try it out.

[117:40] 发言人 1: And if it’s good, I keep it. If it’s not good, I throw it away. And I do this many times.

[117:45] 发言人 1: So the design of their algorithm previously are basically mostly manually done. A human engineer，a human，a human intelligence，now become something that you can try all the time。Just like an agentic system, that’s another way.

[118:04] 发言人 1: Okay, and in the end they were actually being able to have a major publication of this. And online be packing is just one example. In this paper what they actually show is this pipeline leads through a mathematical discovery.

[118:20] 发言人 1: That’s why you see the title is actually mathematical discovery. So this pipeline is actually helped them to develop something. On this mathematical problem, which previously think that’s actually okay.

[118:34] 发言人 1: So in the end, by the end of twenty twenty-three, they have a nature paper now. It becomes a mainstream—not a mainstream—but at least a popular way. You must have heard about how nowadays on the news someone have discovered a new really hard math problem.

[118:50] 发言人 1: Some people use this to design a new drug; some people use this to develop a new something. The underlying pipeline is exactly you want to. Convert a design problem in our class, designing a heuristic function into another search problem where in the search problem largely drive the search.

[119:12] 发言人 1: Okay, so in previous couple of years starting from2023, we have been giving projects on using AI to design AI algorithms. Okay，this will be amazing for this semester as well，so you actually can try using AI writing an AI algorithm or writing a search on top of a large language model to develop an AI algorithm yourself。Okay，so this concludes everything I want to cover for the lecture materials，so before I. Talk about the tutorial, any quick questions?

[120:01] 发言人 1: Okay，if not，give me a second to open this week’s tutorial。

[120:40] 发言人 1: Okay, so if you open your. Let me see the canvas page. We have providing this week’s tutorial in two format. One is in the Google Cloud, if you are manage to you go to Google. If not, this is a local copy, which is. You can render it on your own computer, okay? You need to install the notebook first, otherwise you can go to the link.

[121:06] 发言人 1: So in here, what you will find is every concept we talk about in this today’s class, in this week’s class, we have an Python implementation, okay? We have things to show you how you can implement a problem, a node, search for our tree and accept us. In the Python，and we also implement the depth first search algorithm，breadth first search algorithm，iterated deepening，uniform cost search algorithm，and greedy search and A* search algorithm。

[121:40] 发言人 1: Okay，and we also implement the traveling in Romania case for you。where you can run your algorithm to see its outcome. ok, so what we want you to do? ok, we are not giving this as assignment, but we do want you to look at yourself.

[121:58] 发言人 1: yes. 我。We cover today in class is the concept of the algorithm, but here is what provide you with the actual implementation of the algorithm, which has a lot of details to it, like how you should maintain the frame, how you implement the cube, etc.

[122:18] 发言人 1: These are the details we omit in class, but. We provide a tutorial as as a code, so. Our hope is you can go over this at your own time.

[122:32] 发言人 1: Try to see if the implementation actually is the same as how these algorithms are covered in class. Then we hope you want different algorithms, the A star, the iterative deepening, the bidirectional, the uniform cost search on this problem, on this traveling in Romania problem. We have already entered for you.

[122:53] 发言人 1: All you have to do is run it. You can compare the outcome yourself. to see if it’s. Something you expect, because in class we tell you what’s the advantage and the disadvantage of each one of these algorithms, like what’s the time complexity, what’s the space complexity.

[123:13] 发言人 1: After you run this, you can check after you link to your server, it actually show you the CPU usage, memory usage, and all these. Okay, you can compare yourself. As you run in the algorithm, to see if this is follow the logic, if it follows the conceptual conclusion of each one of these algorithms.

[123:34] 发言人 1: Okay, again, we have implement everything for you so you don’t have to do it, but we do this for our good heart to hope that you can run it. Because you don’t have to implement all the office, you run it and you see it yourself. Whether what we show you in the class is actually happen in the code.

[123:55] 发言人 1: Okay, if you can run it on Google Cloud, nothing is very easy. All you have to do. Place this wrong angle，but you cannot as a Google，then you provide a.

[124:10] 发言人 1: local copy that’s exactly the same as on the google class, and you need to run it on your computer as google as the google notebook. okay, neither one is fine, nothing to submit. we just want you to use your own time and try to get some.

[124:25] 发言人 1: Okay，any question on today’s on this week’s? Okay，if not, you are，you are free to go. That’s it for this week’s lecture，and I’m here for question you may have. Okay.

[124:51] 发言人 1: 后面说的什么？嗯，后面这个它说 material，它说了什么东西？呃。如果有疑问就没有了，可以走了吧。我觉得这个千问的这个做密码 APP 还不错，千问吗？那这个要破课吗？不用，对，破课。